

Rotation of Figures

COMPLETE LEARNING GUIDE

What You'll Learn

Rotation-of-figures questions test whether a candidate can precisely track a figure's angle and direction of turn — determining how much and which way a figure has rotated, telling a genuine rotation apart from a mirror image that merely looks similar, and applying a confirmed rotation step forward through a sequence or to a missing figure. This sub-module builds a reference-feature method for tracking rotation reliably: measuring angle and direction from one distinctive feature, using the turn-sequence test to catch mirror images, rotating figures with off-center internal markings and multi-part compound shapes correctly, and confirming a multi-step sequence's rotation angle against three or more figures rather than just two.

1. Reading Rotation Direction and Angle

Every rotation question specifies (or implies from a shown pair) a direction — clockwise or anticlockwise — and an angle; determine both explicitly by tracking one distinctive point or feature of the figure from its starting position to its new position, and measuring how far around the circle it swept, in which direction.

- A figure's single marked corner starts pointing due north (straight up). After rotation, that same corner points due east (directly right). Moving from north to east sweeps 90° in the CLOCKWISE direction (north ! east is clockwise; north ! west would have been anticlockwise for the same 90°).

2. Verifying a True Rotation vs. a Mirror Image

A figure that has been rotated preserves its exact internal left/right handedness — tracing its outline in one direction produces the same sequence of turns before and after rotation; a mirror image instead reverses that turn sequence, which is the definitive test for telling a genuine rotation apart from a mirror image that merely looks similar.

- An L-shaped figure's outline, traced clockwise from its longest edge, goes: straight, turn-right, straight, turn-left, straight, turn-right (back to start). A candidate 'rotated' version, traced the same way, gives: straight, turn-LEFT, straight, turn-RIGHT, straight, turn-left — the reversed sequence. This candidate is a mirror image, not a valid rotation, despite matching angles and side lengths.

3. Rotating Figures With Internal Asymmetric Markings

When a figure has an internal marking placed off-center, that marking must rotate together with the entire outer figure by the identical angle and direction — check that the marking's position relative to the outer shape's reference feature stays constant before and after rotation.

- A square has a small arrow inside pointing toward its top-right corner. After a 90° clockwise rotation of the whole square, the arrow must now point toward what was the top-LEFT corner before rotation (since that corner has rotated into the top-right position) — the arrow's relationship to 'the corner nearest it' stays fixed; its compass-direction label changes.

4. Multi-Step Rotation Sequences

A sequence showing a figure at 3 or more stages, each rotated a fixed amount further than the last, is solved by first confirming the SAME fixed angle and direction applies between every consecutive pair shown, then extending that same fixed step to find the next figure in the sequence.

- A sequence shows a figure at 0° , then rotated to 60° , then to 120° (each step $+60^\circ$ clockwise, confirmed by checking both gaps in the sequence). The next figure in the sequence must be at 180° — the same figure, rotated a further 60° clockwise from the 120° position.

5. Finding the Odd Figure via Inconsistent Rotation

In a set of figures that are supposed to all be the same base shape at different rotations, the odd one out is typically the one that is either a mirror image (reversed turn sequence) or rotated by an angle that doesn't match the clean, consistent rotation pattern used by the rest of the set.

- Four figures are the same arrow shape at rotations of 0° , 90° , 180° , and 270° (clean 90° steps). A fifth figure appears to be the same arrow, but careful angle measurement shows it's actually at 100° , not a clean 90° -multiple — this figure is the odd one out, since it breaks the consistent rotation-step pattern even though it looks superficially close to the 90° position.

6. Rotation of Non-Symmetric Compound Figures

For compound figures made of multiple distinct parts, rotate the ENTIRE compound as one rigid unit — the attachment point and relative angle between the two parts must stay exactly fixed throughout the rotation, since rotating the parts independently would produce an invalid, physically incorrect result.

- A compound figure has a small triangle attached to the top of a circle, pointing straight up. After rotating the whole compound 90° clockwise, the triangle must now point directly right (attached at the same relative point on the circle's edge, which has itself moved 90° clockwise) — not still pointing up, and not detached from its original attachment point.

7. Using a Fixed Reference Point to Track Rotation

Before rotating any figure mentally, pick one unambiguous reference feature (the longest side, a uniquely shaped protrusion, a single marked vertex) and track **ONLY** that feature's new position and direction — attempting to track the entire figure's overall 'look' at once is unreliable and leads to errors, especially for complex or near-symmetric shapes.

- A five-sided figure has four similar-looking sides and one distinctly shorter fifth side. Using that short side as the sole reference point, track where it ends up after the stated rotation, then reconstruct the rest of the figure's expected position relative to it — far more reliable than trying to track all 5 sides simultaneously.

8. Worked Full Example: Confirming a Rotation Across a Full Set

Combine the full method: identify the reference feature, measure the angle and direction between the first two figures using that reference, verify the same angle and direction holds for a third given figure, and only then apply it forward to find a missing or next figure.

- A reference notch on a hexagon starts at the top (12 o'clock position). In the second figure, the notch is at the 4 o'clock position (120° clockwise). In a third given figure, the notch is at the 8 o'clock position (a further 120° clockwise, confirming the consistent +120° clockwise step). Applying the same step forward: the next figure's notch must be back at the 12 o'clock position (a full 360° cycle completed).

Common Pitfalls

- Judging a rotated figure by overall visual similarity instead of tracking one exact reference feature's angle and direction.
- Mistaking a mirror image for a valid rotation because angles and side lengths match, without checking the turn sequence.
- Forgetting that an internal marking must rotate together with the outer shape by the identical angle and direction.
- Assuming a multi-step rotation sequence's angle from only the first two figures without checking it against a third given figure.
- In compound figures, rotating the individual parts independently instead of rotating the whole rigid unit together.
- Mistaking a figure rotated by a close-but-not-exact angle as fitting a consistent rotation-step pattern.
- Losing track of direction (clockwise vs anticlockwise) partway through a multi-step sequence.

Key Takeaways

- Track one clear reference feature's exact angle and direction of movement rather than judging rotation by overall visual impression.

- A true rotation preserves the figure's left/right turn sequence; a mirror image reverses it — this is the definitive test.
- Internal markings rotate together with the outer shape by the identical angle and direction, not independently.
- Confirm a multi-step sequence's fixed rotation angle against at least three consecutive figures, not just the first two.
- Rotate compound figures as one single rigid unit, keeping attachment points and relative angles between parts fixed.
- The odd figure in a rotation set is often a mirror image, or an angle that's close to but not exactly on the consistent rotation step.

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